NC STATE UNIVERSITY

NS-3 Code Bases

Zifan Zhang
Networking & IntelligenCE (NICE) Lab

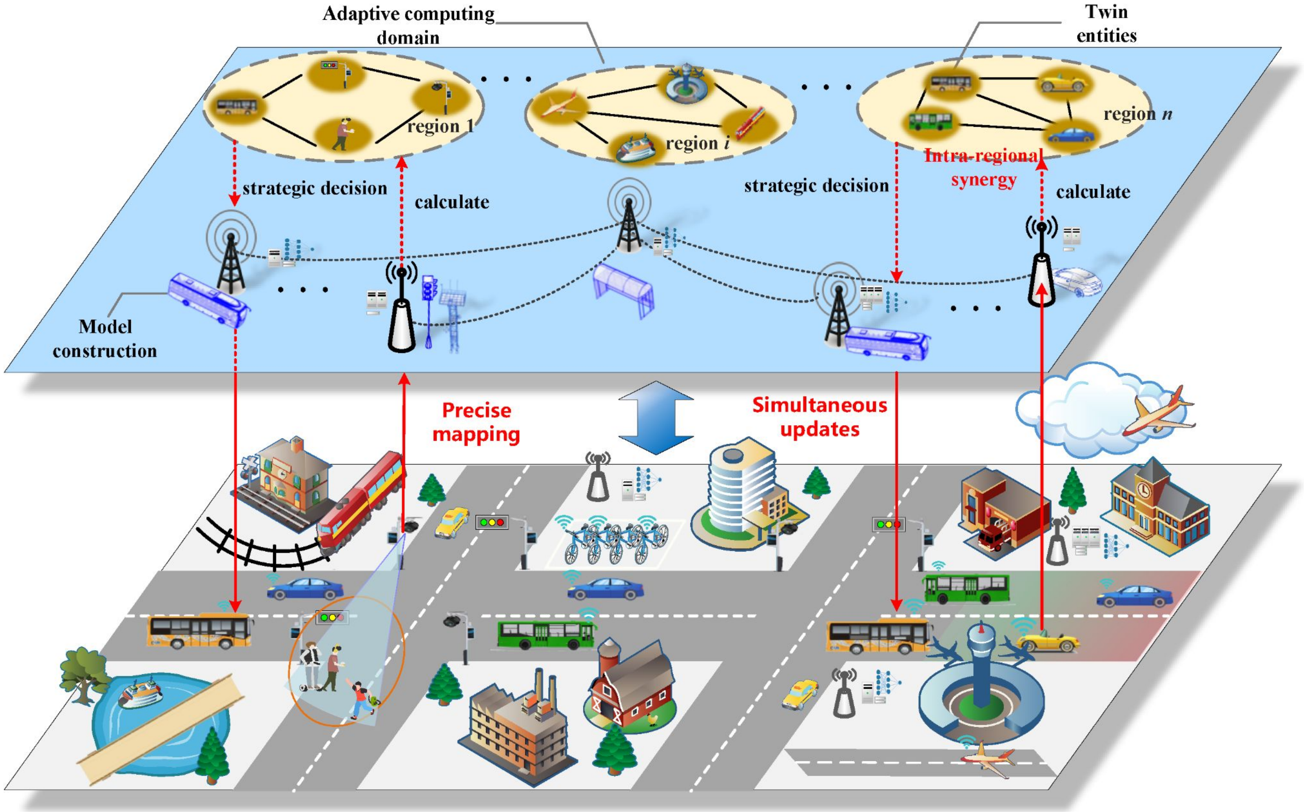
https://ns3.netsimsolutions.com/catalog

Why are Network Important?

1. **Stay connected 24/7** – texting, video-calls, social media.
2. **Instant information & entertainment** – YouTube, TikTok, game updates, music streaming.
3. **Smart devices all around us** – watches, speakers, thermostats (Internet of Things).
4. **Life-saving technology** – emergency calls, remote medical care, disaster alerts.
5. **Future lifes** – satellite, cybersecurity, robotics, space Internet.

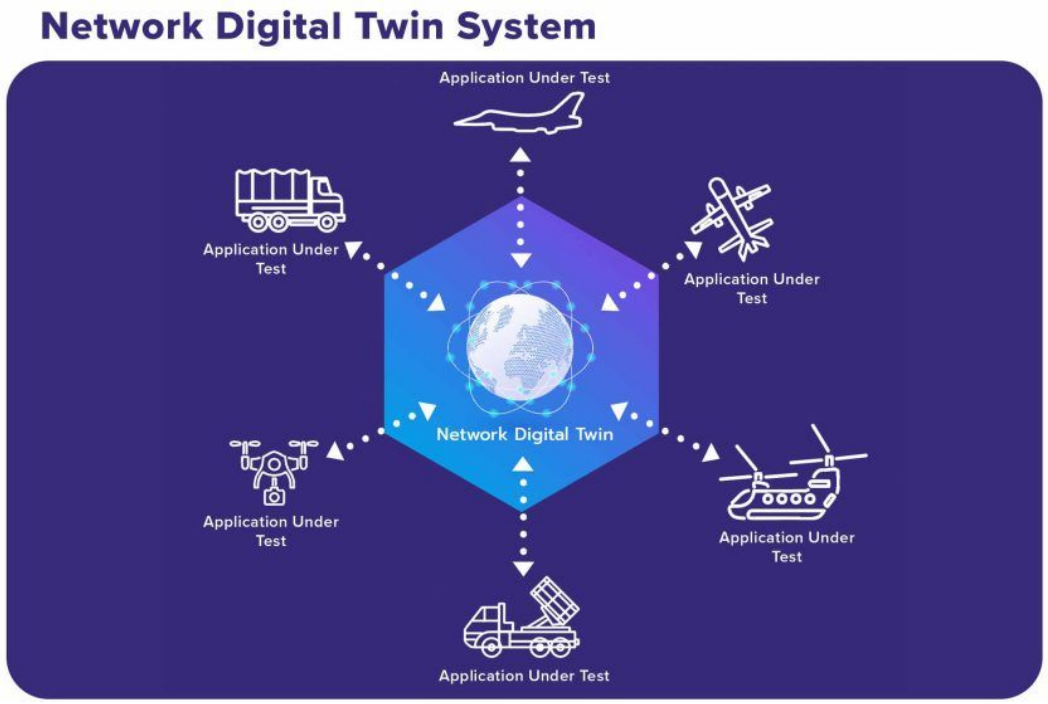
Real Testbed vs. Simulation Platform

Testings and validations on real testbed may not be cost-effective. More factors, such as safety risks and scalability, need to be considered in real testbeds.



Digital Twins vs. Simulation Platform

Digital twins are live replicas of physical network system. Simulator stays in sync with sensors/metrics from the physical system.



Code Bases

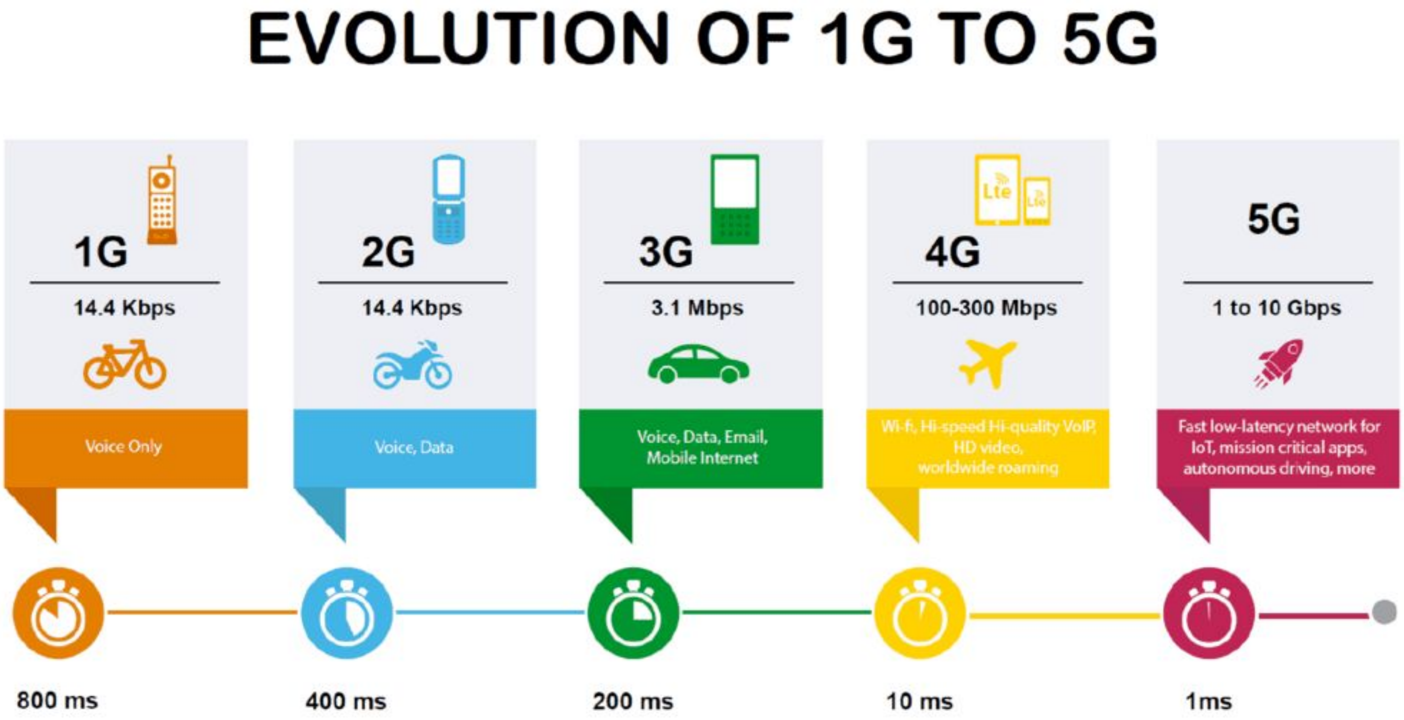
1. **ns-3 Core (3.45) –** the engine room; models Ethernet & Wi-Fi 7 with new DHCPv6 and 320 MHz channels
2. **5G-NR (5G-LENA 4.0) –** builds city-scale 5G cells, beam-forming & ultra-low-latency demos; fresh release for MWC 25
3. **NR V2X (v2x-1.1) –** lets cars chatter wirelessly for crash warnings and platooning; adds 5G sidelink modes 3 & 4
4. **LoRaWAN (0.3.2) –** ultra-low-power radios for smart farms & air-quality sensors; runs for years on a coin cell
5. **ns3-O-RAN (1.3) –** simulation of an open RAN; includes a Near-RT RIC that tweaks network slices in real time
6. **LEO-Sat Module (3.43) –** simulates Starlink-style constellations and hand-overs at 27 000 km/h

Where to Find them

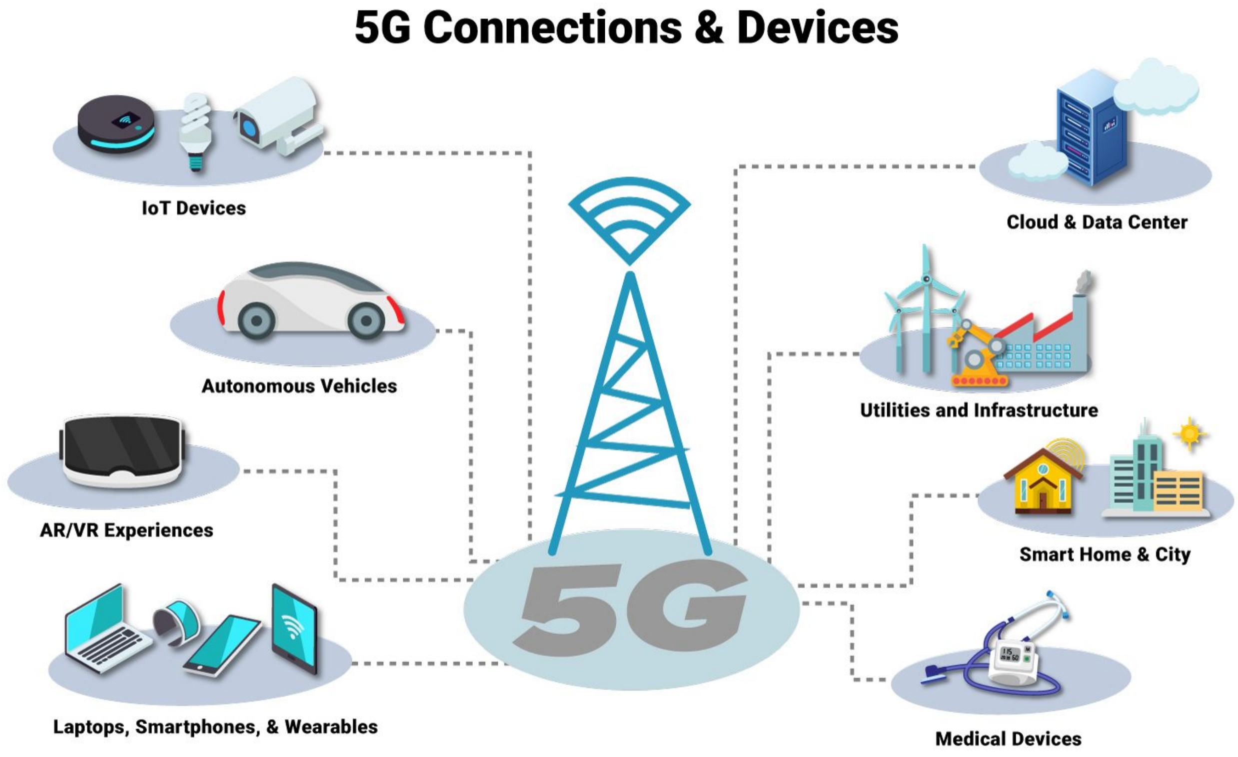
Select the ns-3 code base from the list below:

Index	Name	Repository	Selection
1	ns-3.44	https://gitlab.com/nsnam/ns-3-dev/-/tree/ns-3.44	◉
2	5g-nr	https://gitlab.com/cttc-lena/nr/-/tree/5g-lena-v4.0.y	○
3	nr v2x	https://gitlab.com/cttc-lena/nr/-/tree/v2x-1.1	○
4	lorawan	https://github.com/signetlabdei/lorawan/tree/v0.3.2	○
5	ns3-oran	https://github.com/usnistgov/ns3-oran/tree/v1.3	○
6	sns3-satellite	https://github.com/sns3/sns3-satellite/tree/3.43	○

Cellular Generations

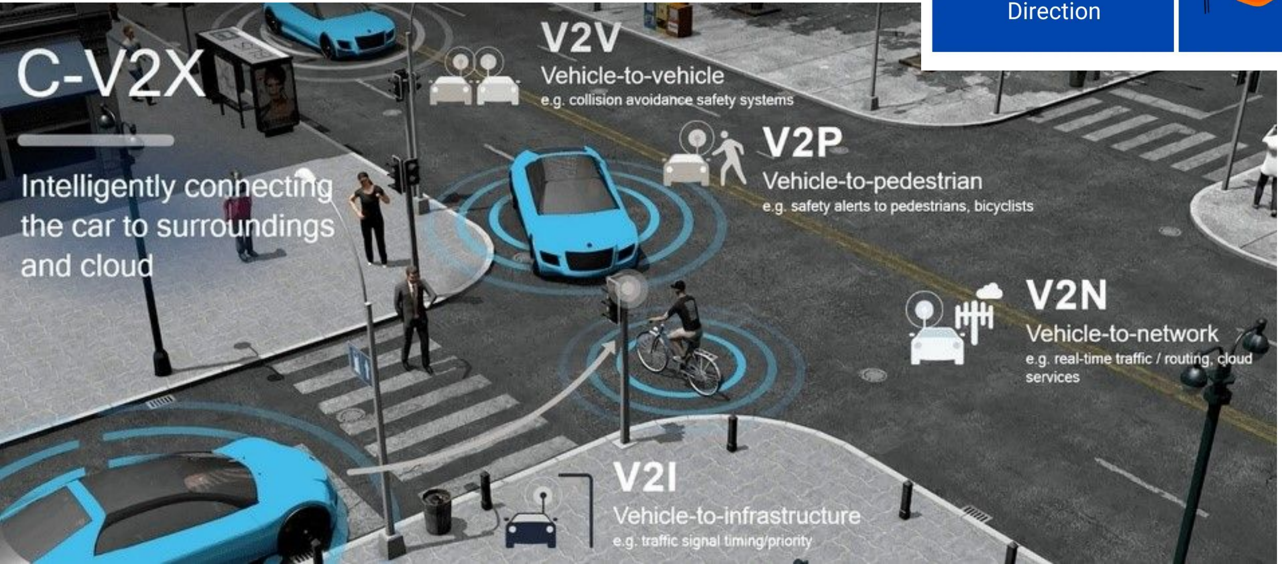


5G Networks Benefits

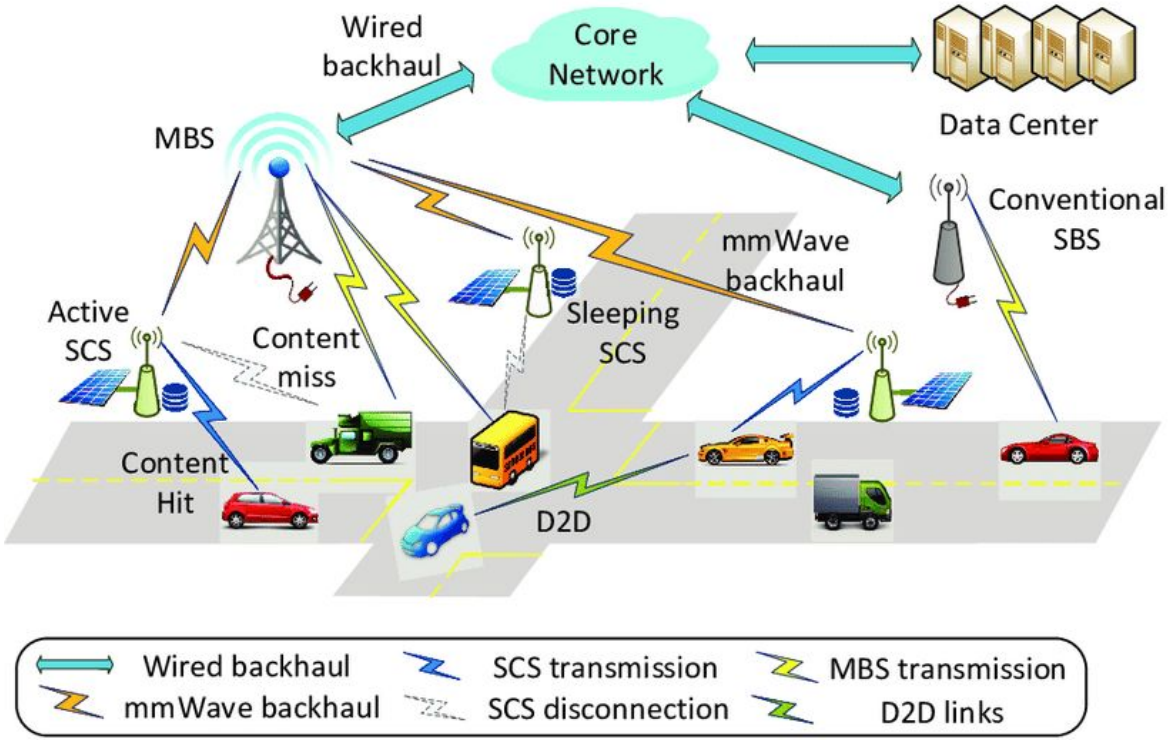


Vehicle to Everything

Vehicles are connected via wireless networks and coordinated by a few cloud servers.



How Vehicular Networks Work

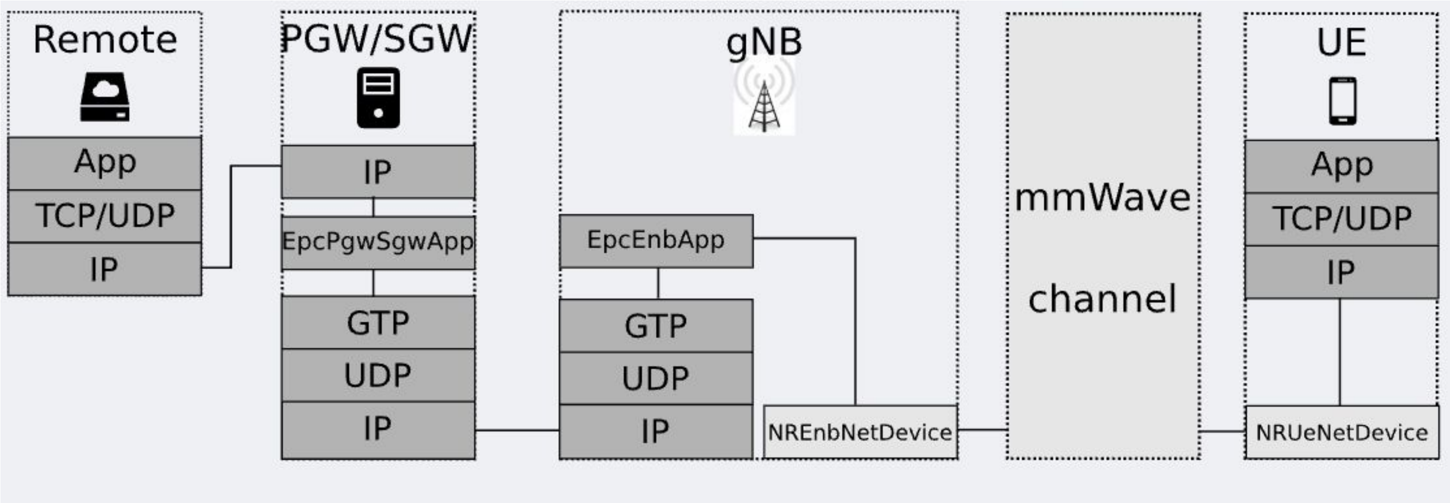


LoRaWAN Architecture

There are three kinds of devices in a LoRaWAN network:

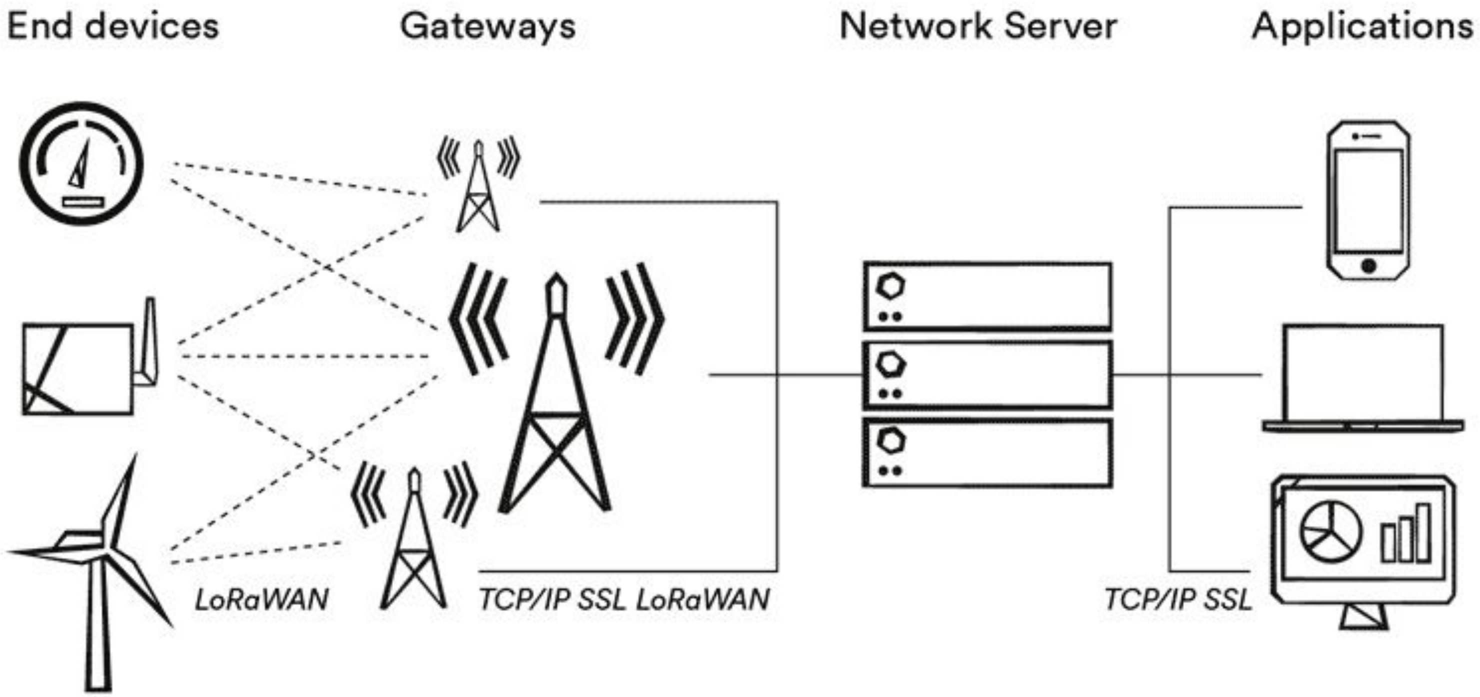
- End Devices (EDs)
  - End Devices are basic network nodes: typically inexpensive, they are constrained by low computational capabilities and are usually powered by a battery.
- Gateways (GWs)
  - Gateways are high-end, mains powered devices that are tasked with collecting the data transmitted by End Devices leveraging the LoRa modulation.
- a Network Server (NS)
  - After a packet is correctly received, it is forwarded to the Network Server via a link with high reliability and speed. The Network Server functions as a sink for data coming from all devices, and as a controller of the network that can leverage some MAC commands to change transmission settings in the End Devices.

Overall Architecture of 5G and V2X Module

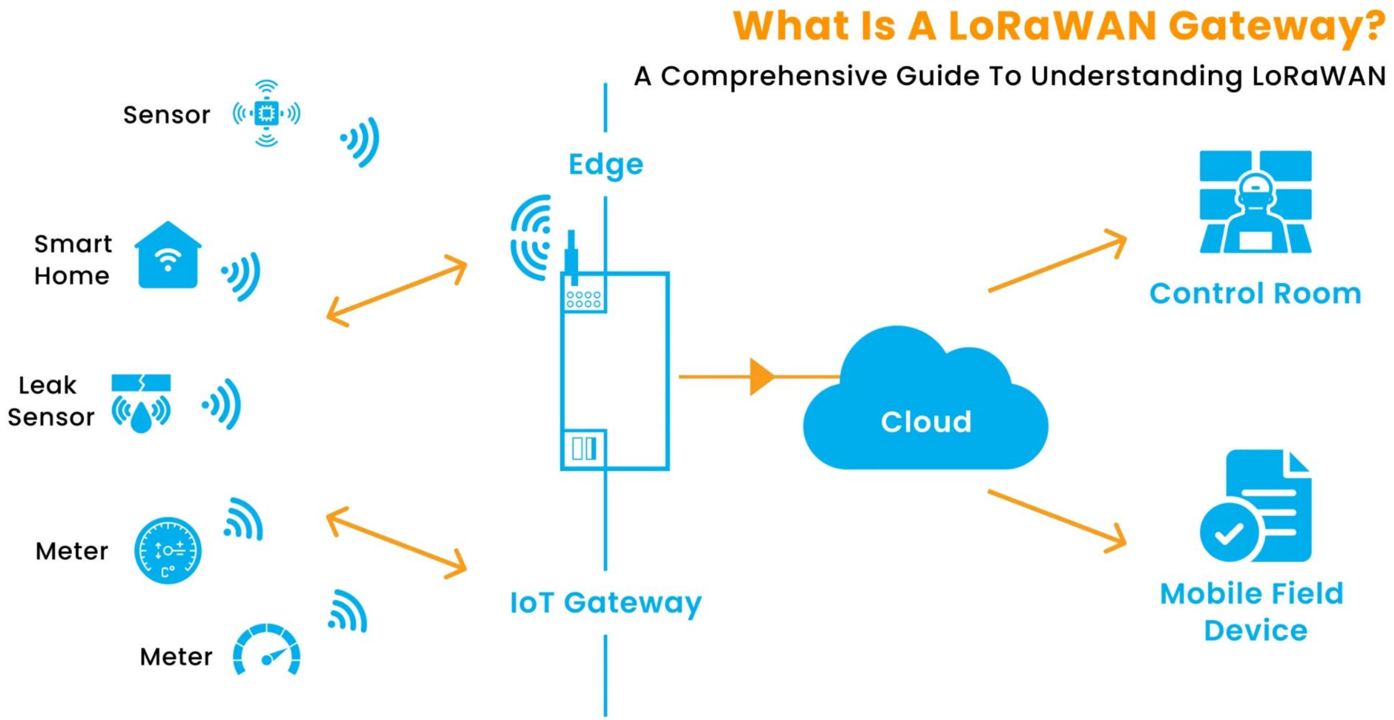


PHY & MAC realism — CSI-RS / CSI-IM measurements, multi-panel antennas, Kronecker & quasi-omni beamforming.
Flexible numerology — sub-6 GHz to 100 GHz with scalable sub-carrier spacing and TDD/FDD.
AI-powered scheduling — Reinforcement-learning TDMA / OFDMA schedulers.
Channel models — 3GPP, NYUSIM, FTR plus new NrChannelHelper for quick swaps.

LoRaWAN Architecture



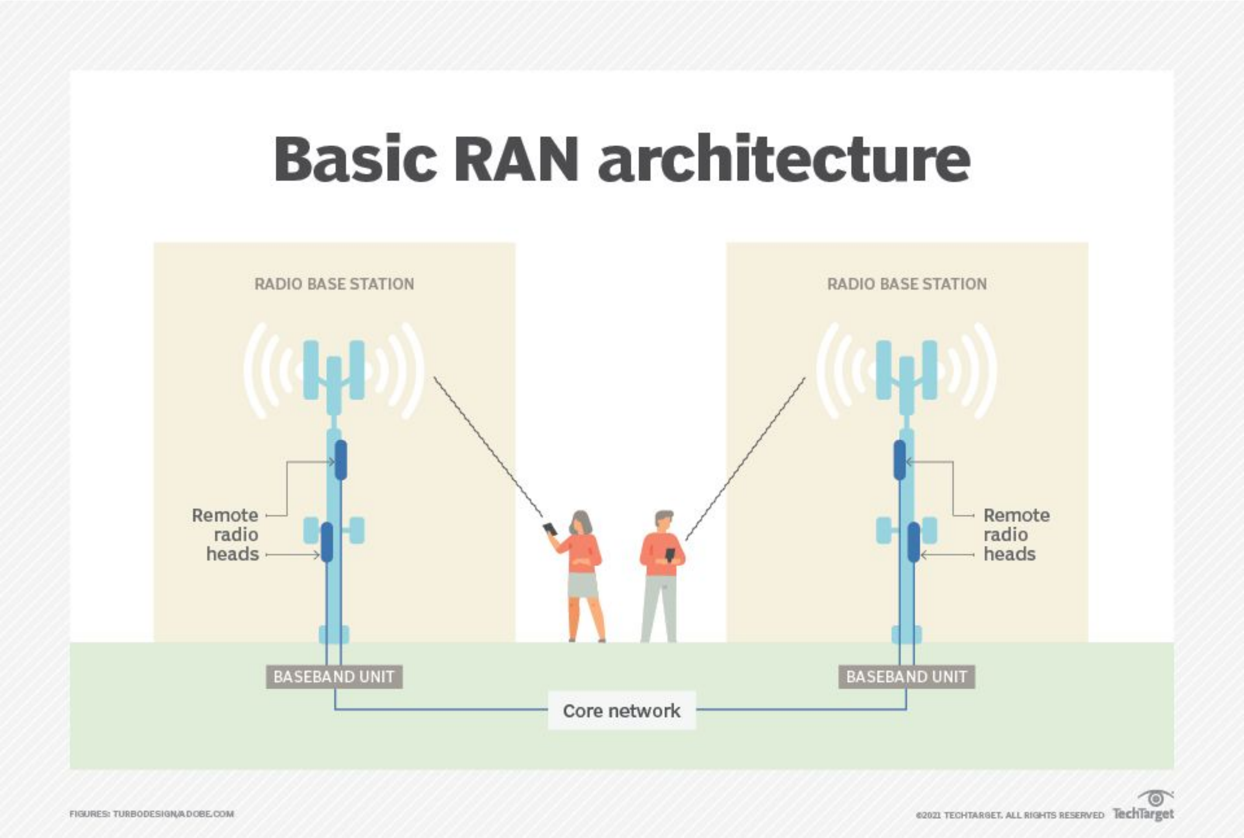
LoRaWAN Architecture



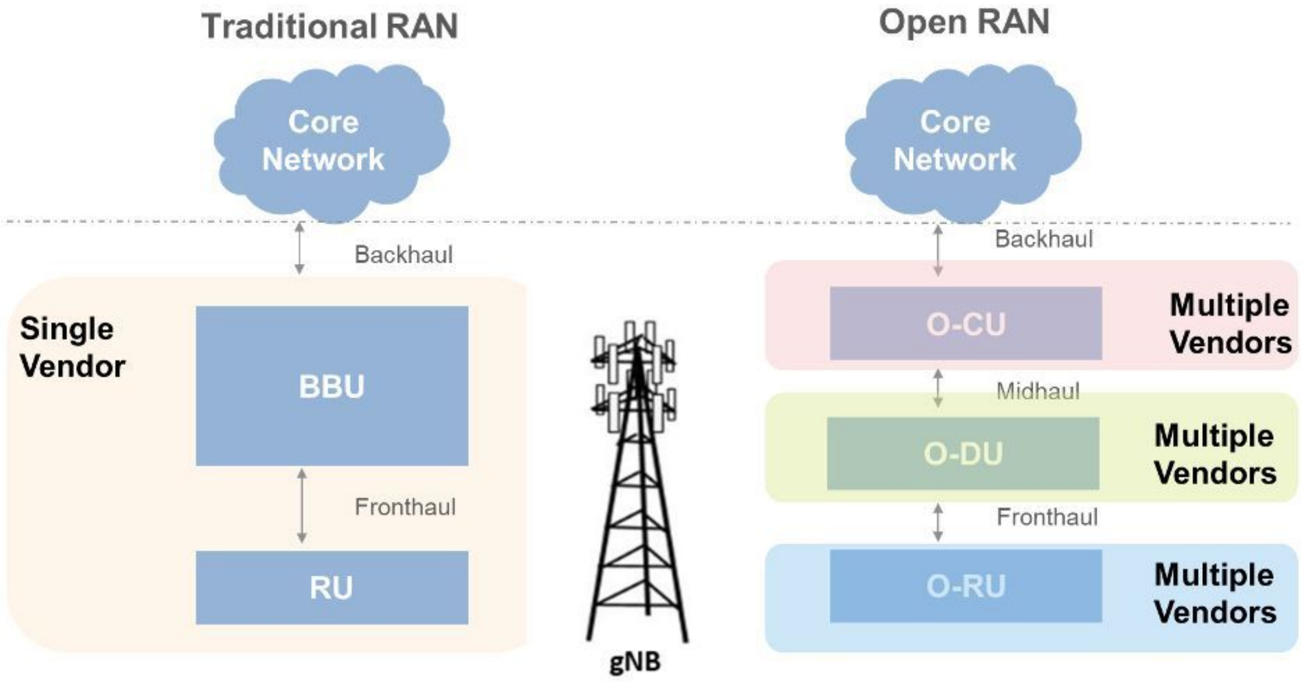
LoRaWAN Benefits

1. **Long-range, low-power links**: 10–15 km in rural areas (2–5 km urban) while squeezing 5-10 year battery life from coin-cell-class power budgets.
2. **Low deployment cost**: cheap sub-GHz radios, license-free ISM spectrum, and a star-of-stars topology that lets one gateway serve thousands of nodes.
3. **Strong security by default**: dual AES-128 layers isolate network control from application data, keeping keys local and blast radius minimal.
4. **Built-in flexibility & scale**: Adaptive Data Rate (ADR) auto-optimizes airtime; networks can be fully private, public, or hybrid with seamless roaming.
5. **Extra features without extra silicon**: gateway-based TDOA geolocation, multicast FOTA, and Class B/C downlinks add tracking and update capability at near-zero power cost.

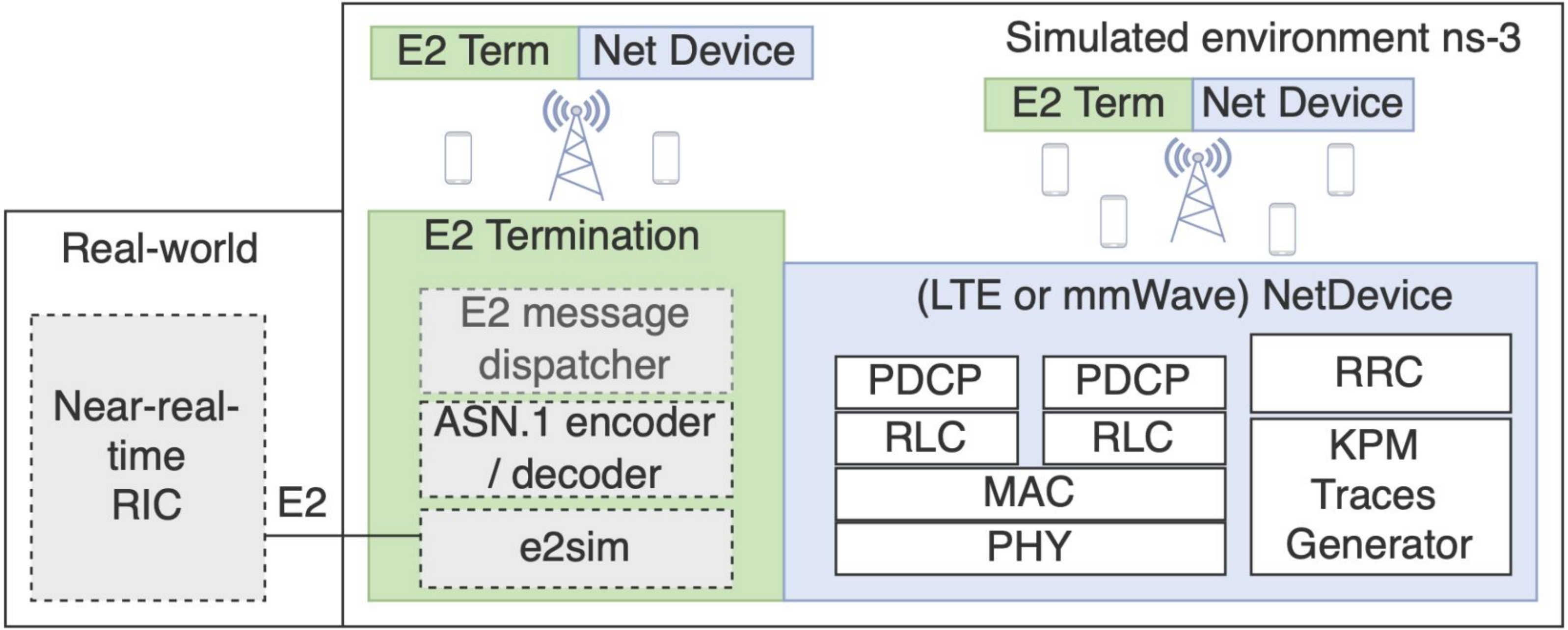
Basic RAN


Basic RAN architecture
RADIO BASE STATION
RADIO BASE STATION
Remote radio heads
Remote radio heads
BASEBAND UNIT
BASEBAND UNIT
Core network
FIGURES: TURBODESIGN/ADOBE.COM
©2021 TECHTARGET. ALL RIGHTS RESERVED
TechTarget

Traditional vs. Open RAN



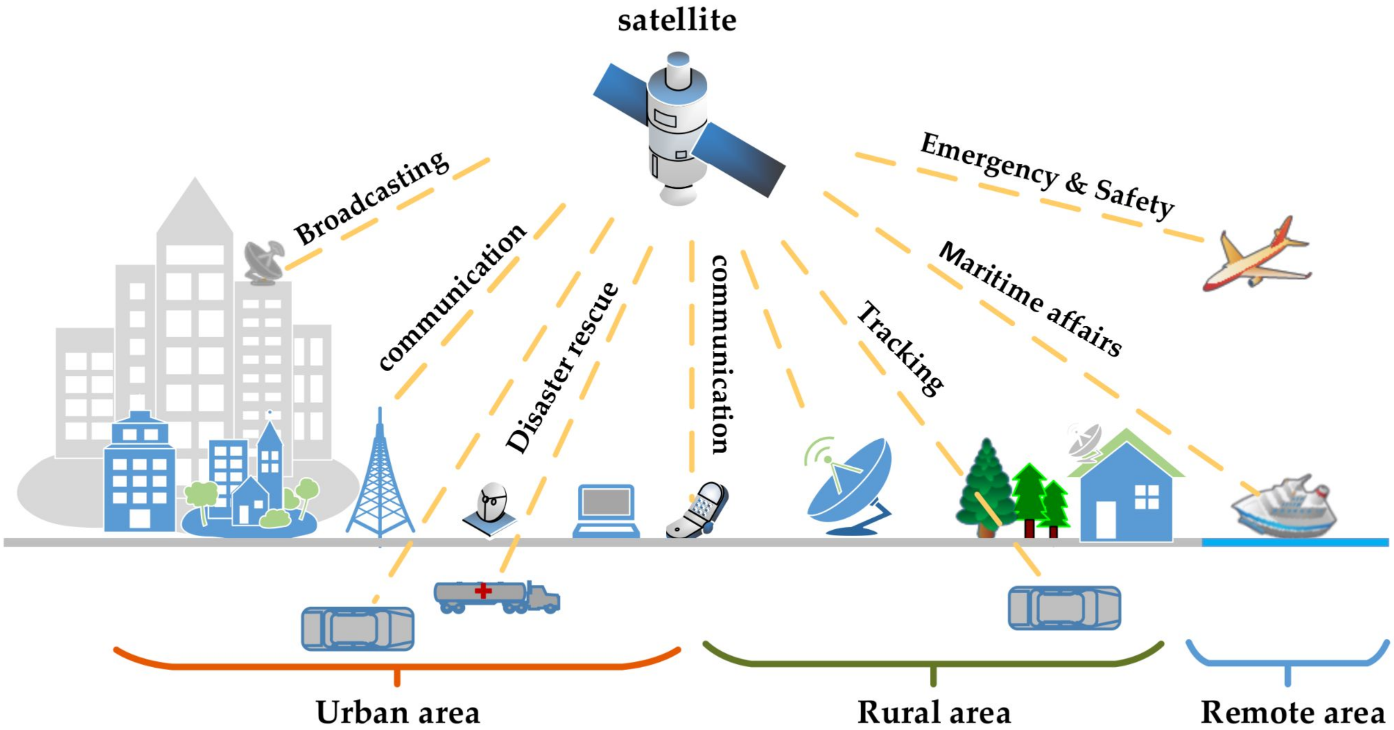
O-RAN Architecture



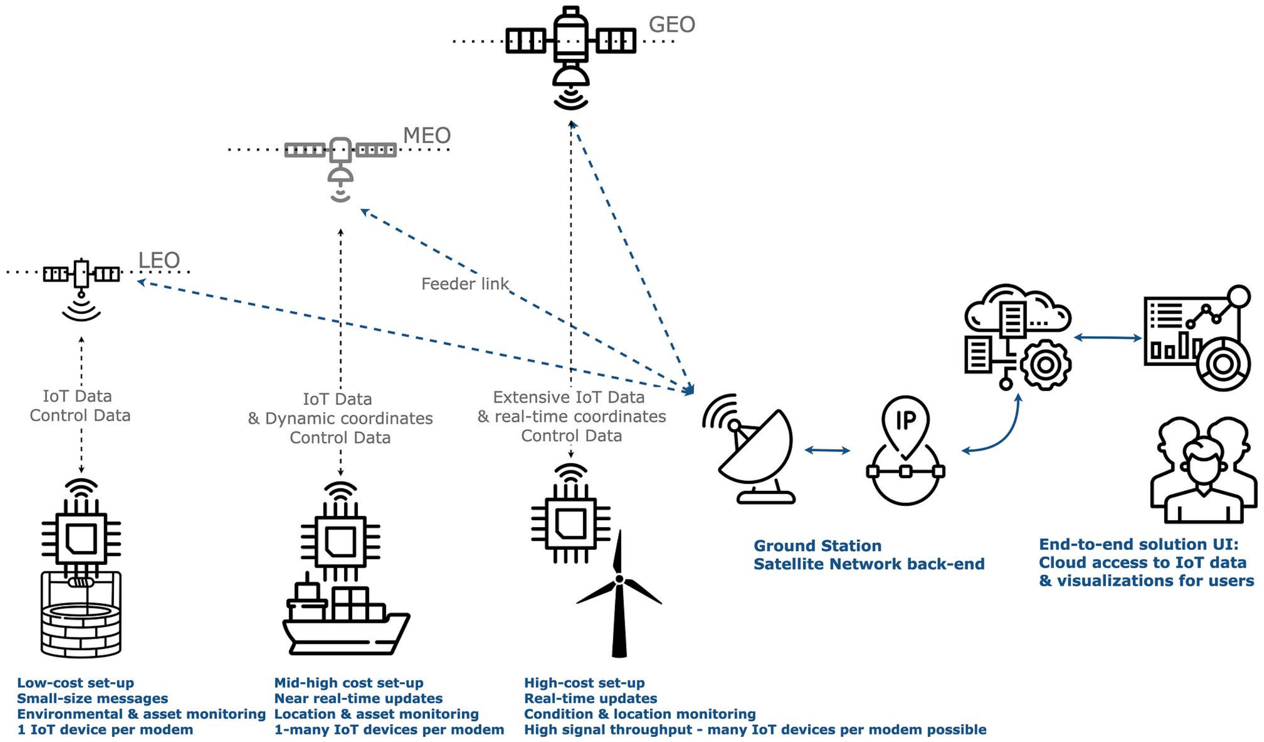
O-RAN Benefits

Vendor-neutral interoperability: Open, 3GPP-aligned interfaces let operators combine radios, baseband, and software from multiple suppliers, ending single-vendor lock-in.
Lower total cost of ownership: Virtualized RAN functions on commercial off-the-shelf servers and competitive component sourcing cut CapEx and OpEx.
Quicker feature roll-outs: Cloud-native design and software upgrades accelerate deployment of 5G/6G capabilities across heterogeneous networks.
Built-in telemetry for AI automation: Standardized RAN Intelligent Controller (RIC) and open O-O1/A1 interfaces expose data for analytics, closed-loop optimisation, and service assurance.
Greener operations: Software-defined baseband allows dynamic sleep modes and workload shifting, improving energy efficiency without specialised hardware.

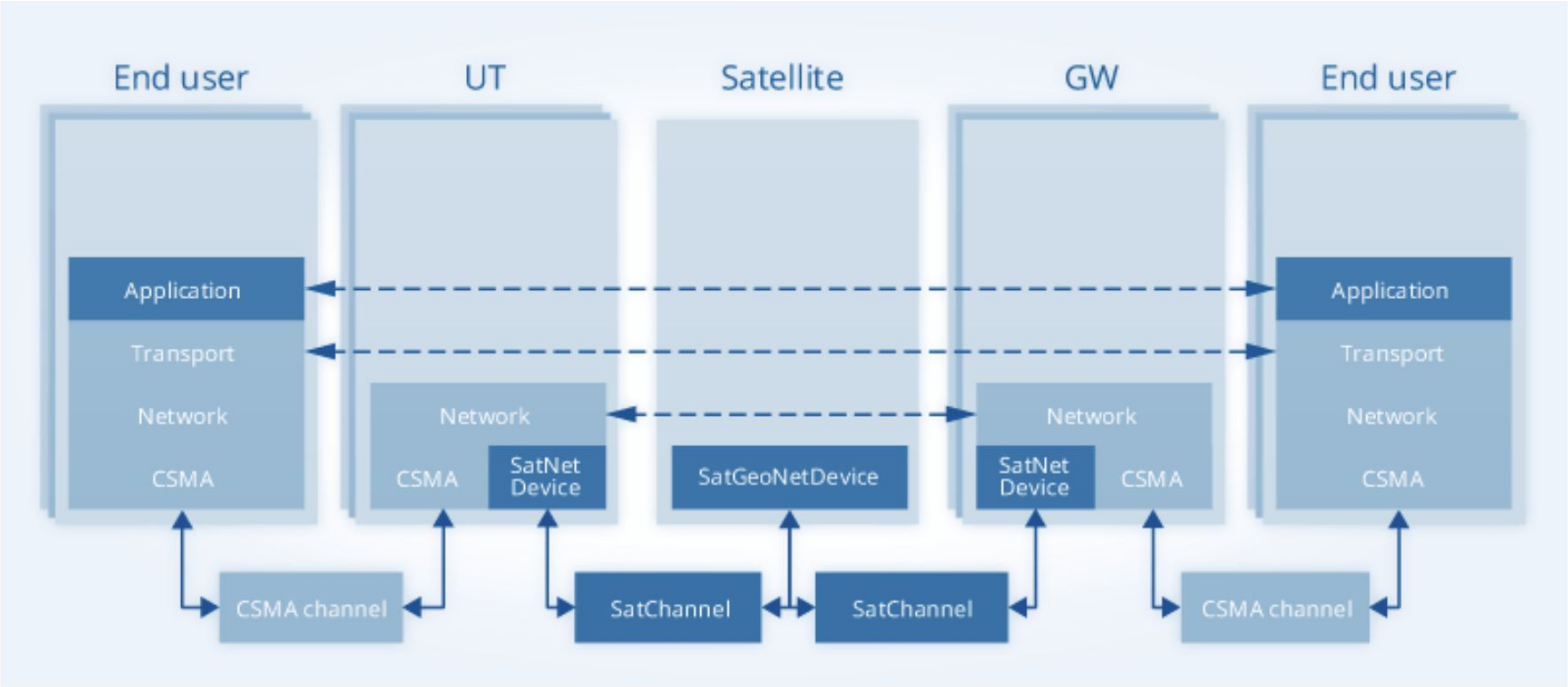
Satellite Networks



Satellite Networks Types



Satellite Network Architecture



NS-3 Local Installation Guide